AWS IAM Operations: Managing Users, Groups, and Roles

Securely Controlling Access to Your AWS Resources

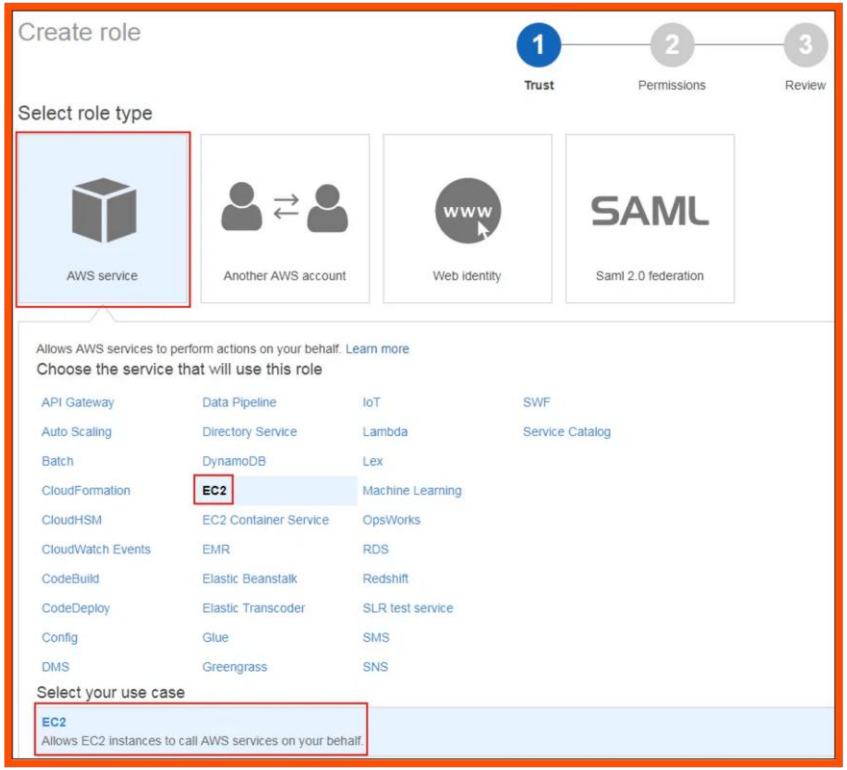
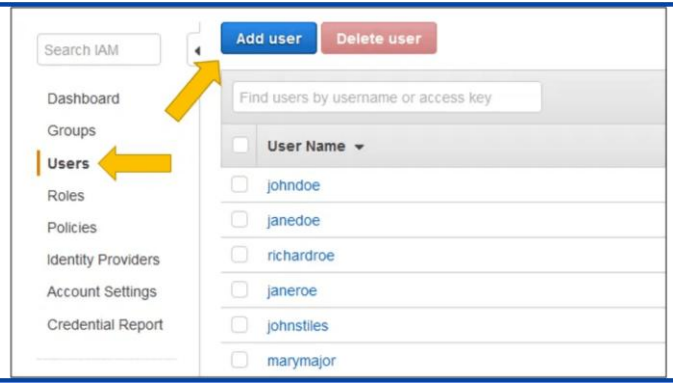


Figure 1. AWS IAM Management Console

What is AWS IAM?

- **Definition:** Identity and Access Management (IAM) is a web service that helps you securely control access to AWS resources.
- **Authentication:** Controls who is authenticated (signed in) to use your AWS resources.
- **Authorization:** Controls who is authorized (has permissions) to use specific resources.



Key Concepts

- **IAM Users:** Individual identities for people or applications.
- **IAM Groups:** Collections of users for easier permission management.
- **IAM Roles:** Identities assumable by trusted entities for temporary access.
- **IAM Policies:** JSON documents that define permissions.

IAM Users: Individual Access Identities

Defining and Securing User Access

- **Definition:** An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.
- **Root vs. IAM:** The root user has full access; IAM users should be used for daily tasks following the **Principle of Least Privilege**
  .
- **Best Practices:** Enforce strong password policies, enable Multi-Factor Authentication (MFA), and rotate access keys regularly.

Figure 3. Managing Users in the IAM Console

Step-by-Step: Creating a New IAM User

Initial Configuration in the AWS Console

- **Navigate to IAM:** Sign in to the AWS Management Console and open the IAM console at **console.aws.amazon.com/iam/** .
- **Add User:** In the navigation pane, choose **Users** and then choose **Add users** .
- **User Details:** Type the user name for the new user. You can add multiple users at once.
- **Access Type:** Select the type of access: **Programmatic access** (for API/CLI) or **AWS Management Console access** .

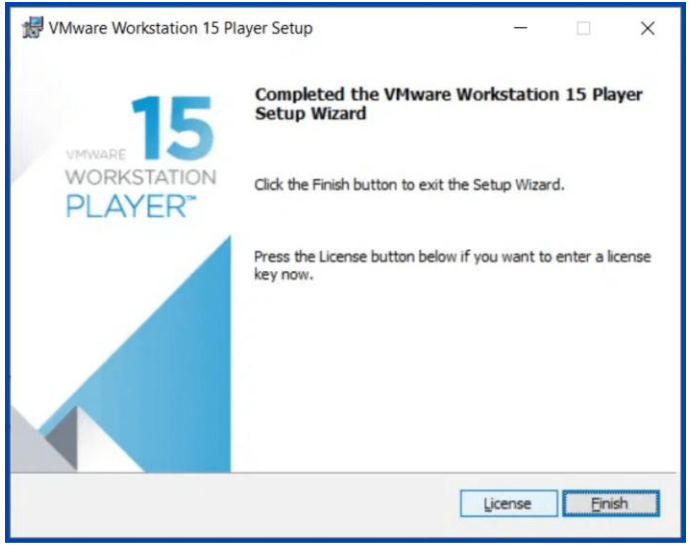


Figure 4. Initiating the Add User Process

Step-by-Step: Setting User Permissions

Assigning Access Control

- **Add User to Group:** The recommended method. Assign permissions to a group and add the user to it for easier management.
- **Copy Permissions:** Inherit permissions from an existing IAM user to ensure consistency across similar roles.
- **Attach Policies Directly:** Assign specific AWS managed or custom JSON policies directly to the individual user.
- **Best Practice:** Always aim for the **Principle of Least Privilege** by granting only the necessary permissions.

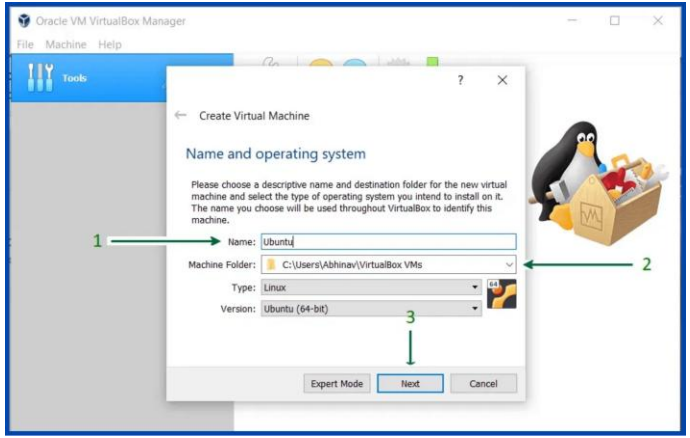


Figure 5. Configuring User Permissions in the AWS Console

Step-by-Step: Review and Create User

Finalizing the User Creation Process

- **Review Details:** Carefully verify the user name, access types, and assigned permissions or group memberships.
- **Create User:** Click the **Create user** button to finalize the identity in your AWS account.
- **Manage Credentials:** For programmatic access, download the **.csv** file containing the Access Key ID and Secret Access Key.
- **Security Note:** This is the only time the Secret Access Key is shown. Store it securely and never share it.

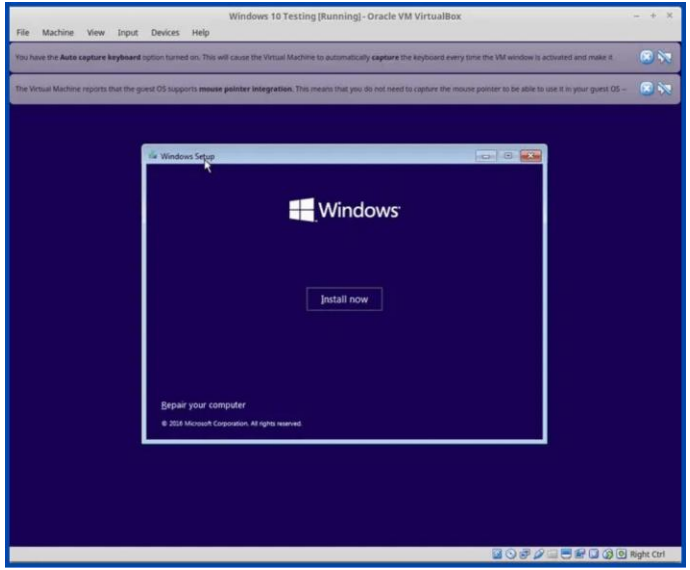


Figure 6. Final Review and Credential Management

IAM Groups: Managing Permissions Efficiently

Simplifying Access Control for Multiple Users

- **Definition:** An IAM group is a collection of IAM users. Groups let you specify permissions for multiple users, making it easier to manage them.
- **Centralized Management:** Instead of attaching policies to individual users, you attach them to a group. Any user added to the group automatically inherits its permissions.
- **Organizational Alignment:** Groups typically represent job functions or departments, such as **Admins**, **Developers**, or **Finance**.
- **Best Practice:** Use groups to assign permissions rather than individual users to ensure consistency and reduce administrative overhead.
- **Scalability:** As your organization grows, managing permissions through groups becomes essential for maintaining a secure and organized AWS environment.

Step-by-Step: Creating a New IAM Group

Organizing Users and Permissions

- **Navigate:** In the IAM navigation pane, choose **User groups** and then choose **Create group**.
- **Group Name:** Type a descriptive name for the group (e.g., "Admins" or "Developers").
- **Add Users:** Select the users you want to add to the group from the list (optional).
- **Attach Policies:** Search for and select the permission policies to attach to the group.
- **Finalize:** Review your selections and choose **Create group** to complete the process.

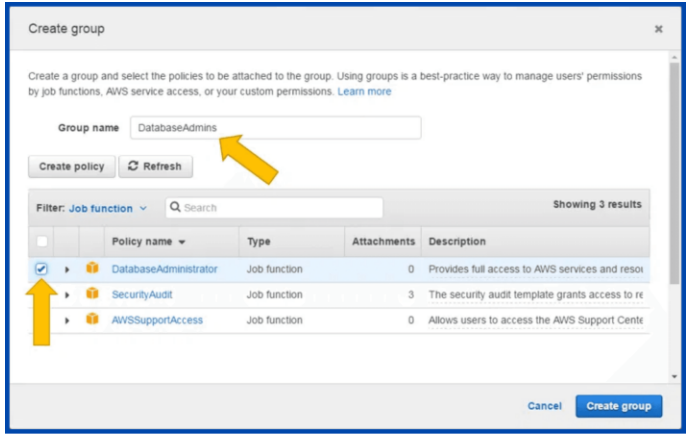


Figure 8. Creating an IAM Group in the AWS Console

IAM Roles: Secure Access for Services and Users

Understanding Temporary Security Credentials

- **Definition:** An IAM role is an identity with specific permissions, similar to a user but not uniquely associated with one person.
- **Temporary Access:** Roles do not have long-term credentials (passwords or access keys). Instead, they provide temporary security credentials.
- **Use Cases:** Granting permissions to AWS services (like EC2 or Lambda), cross-account access, and federated user access.
- **Best Practice:** Use roles for applications running on EC2 instances to avoid storing long-term credentials on the instance.

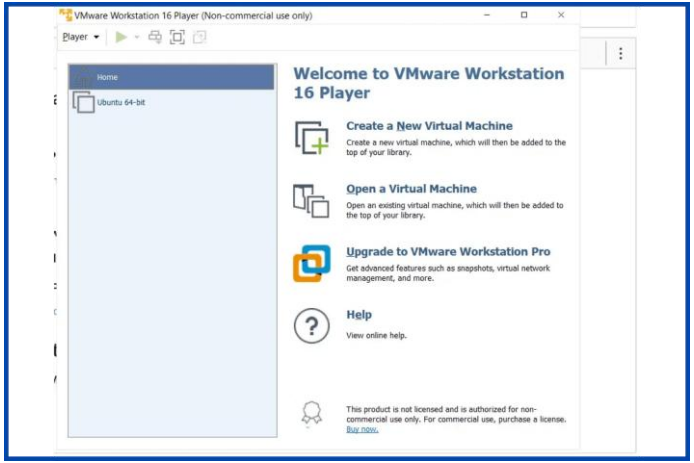


Figure 9. IAM Roles and Trusted Entities

Step-by-Step: Creating a New IAM Role

Defining Trust and Service Access

- **Navigate:** In the IAM navigation pane, choose **Roles** and then choose **Create role**.
- **Trusted Entity:** Select the type of trusted entity. Common choices include **AWS service** (like EC2 or Lambda) or **Another AWS account**.
- **Select Service:** Choose the specific AWS service that will assume this role to perform actions on your behalf.
- **Use Case:** Select the specific use case for the service (e.g., "EC2 allows EC2 instances to call AWS services").

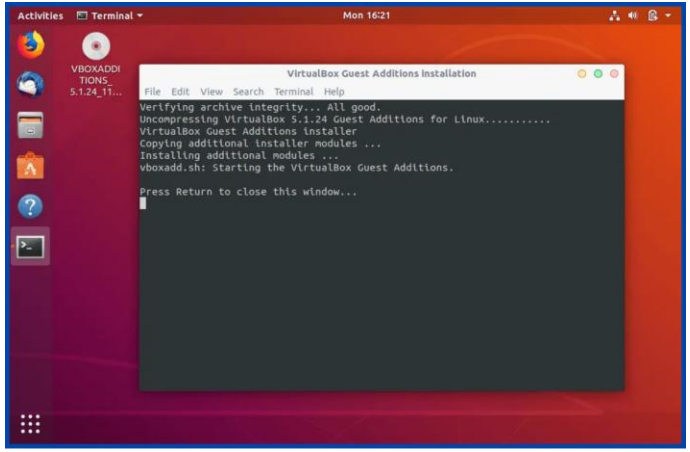


Figure 10. Selecting Trusted Entities for a New Role

Step-by-Step: Attaching Policies and Review

Finalizing the IAM Role Creation

- **Attach Permissions:** Select the policies that define what the entity assuming the role is allowed to do. You can use AWS managed policies or create custom ones.
- **Add Tags:** (Optional) Add metadata to the role using key-value pairs to help organize, track, or control access to the role.
- **Name and Review:** Provide a descriptive **Role name** and an optional description. Review the trusted entities and permissions.
- **Create Role:** Click **Create role**. The role is now ready to be assumed by the specified AWS service or identity.
- **Verification:** Locate the new role in the Roles list to view its **ARN** (Amazon Resource Name) for use in your configurations.

Conclusion: IAM for Cloud Security

- **Core Identities:** We have covered the management of **Users** (individual access), **Groups** (efficient permission management), and **Roles** (temporary, secure access for services).
- **Principle of Least Privilege:** Always grant only the minimum permissions required for a task to reduce the potential impact of security incidents.
- **Multi-Factor Authentication (MFA):** Enforce MFA for all users, especially those with administrative privileges, to add an essential layer of security.
- **Regular Auditing:** Periodically review IAM users, groups, and roles to remove unused identities and ensure permissions remain appropriate.
- **Security First:** Effective IAM management is the foundation of a secure AWS cloud infrastructure.